

Effect of debris mixing on ignition and propagation of dust clouds in fine rice husk

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Highlights

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Crushed brown rice substantially enhances the dispersibility of the dust cloud.

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The addition of crushed brown rice caused the acceleration of the FSV.

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Adding crushed brown rice will reduce the MIT of dust clouds.

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Adding crushed brown rice aggravates the risk of dust cloud ignition and spread.

Abstract

As the associated product of [rice husk](#) dust, large particles of crushed brown rice (abbreviated as debris) are widely found in grain dust processing and transportation industry. The [rice husk](#) dust is combined with huge pieces of debris, which can reduce the consequences of dust explosion in the mixed dust cloud. Debris added at a 70 % ratio considerably slows the spread of flames. However, at 500 g/m³ of dust cloud concentration, including a 10 % percentage of debris increases the average flame spread velocity (AFSV) by 16.9 % and the maximum flame spread velocity (MFSV) by 33.3 %. This intensify phenomenon persists across varying dust cloud concentrations. Tests on the [dispersibility](#) of blended dust clouds revealed that fine dust particles that have been adsorbed on top of bigger particles typically stripped off upon lifted, increasing the blended dust cloud's [dispersibility](#). This leads to the formation of a dust cloud system with

efficiently smaller particle sizes, consequently resulting in an increase in flame spread velocity (FSV). Moreover, owing to the enhanced dispersibility of the dust cloud due to the inclusion of larger particles, the minimum [ignition temperature](#) (MIT) of rice husk dust clouds decreases. Analysis of the angle of repose (AOR) and Hausner ratio (HR) indicates that the incorporation of larger particles improves the flowability of the blended dust cloud, thereby increasing the risk of dust explosions.

Introduction

Combustible dust is one of the hazard sources in the process industry, with fire and explosion risk (Amyotte et al., 2007, 2012; Amyotte, 2006, 2014). The dust explosion occurs widely in the processing and transportation industry (Chen et al., 2018; Zhang et al., 2024a). In the grain dust transportation and processing process industry, dust explosion accidents resulting in casualties have been frequently reported. For instance, the granary in Turkey was the scene of a dust explosion catastrophe in 2023. The movement of wheat and wheat dust from the ship to the granary was the source of the explosion. Large particles and tiny dust coexisted in an explosive atmosphere in these events (Bellochio and Coradi, 2024). Therefore, in order to realize the more effective process safety in the dust transportation industry, it is of great practical significance to carry out the research on the dust explosion protection technology in the grain processing and transportation industry.

Existing research on the dust explosion-proof measures in the grain dust processing and transportation process is mainly based on the fire and explosion characteristics of pure fine dust (Meng et al., 2022a; Li et al., 2021, 2023; Yuan et al., 2023; Zhang et al., 2024b, 2024c; Sun et al., 2024; Sahu and Mishra, 2022). Overlooking the ignition and explosion differences caused by the interaction between large particles and fine dust. For instance, the risks associated with explosions and fires in pipelines where debris and fine rice husk dust are combined are frequently overlooked in the transportation and producing of rice. Due to the influence of processing techniques or the materials themselves, large particles are rarely found in isolation during vertical mechanical transportation processes. In the process of grain production and processing, the particle diameter of crushed brown rice is about 450 μm , but the particle diameter of fine rice husk is only about 20 μm .

Previous studies have revealed that the blending of sizable particles can possess an impact akin to that of inert substances. The probability of explosion is much decreased by the incorporation of coarse particles, highlighting the significance of coarse particles in averting dust explosions (Man and Harris, 2014). Furthermore, larger Al_2O_3 particles have less of an inhibiting influence on the ignition of Al dust clouds than do finer inert powders (Bu et al., 2021). However, there are also some anomalous cases where the addition of large particles can actually exacerbate the possibility of dust explosion. Existing research

has found that this is mainly due to the improved dispersibility of the blended dust cloud caused by the coarse particles. Large particles have a similar effect to powder anti-caking agents, causing small particles to transition from clustering with each other to adhering to the surface of large particles, and subsequently detaching from the surface under the dispersing turbulent action, raising the dust cloud's percentage of fine particles (Perera et al., 2019; Marmo et al., 2019). Further studies by Eckhoff (2012), Eckhoff (2013), have shown that under the same intensity of turbulent action, the addition of powder anti-caking agents such as SiO_2 can alter the van der Waals forces between micron-sized particle clusters, resulting in finer dispersion of particle sizes and higher Kst values. These research findings provide a new mechanism for the finer dispersion of dust leading to increased risks of dust explosion. Thus, most research efforts, both domestically and abroad, have focused on the phenomenon of typical tiny dust particles igniting and exploding (Yang et al., 2024; Li et al., 2016; Bu et al., 2020a). Rarely addressed are the operational scenarios in actual production settings involving the more prevalent inclusion of large particulate matter (such as crushed brown rice) and its concomitant mingling with fine dust during vertical transportation processes. Disregarding this phenomenon may lead to deficiencies in dust explosion detection and warning, as well as the ineffectiveness of venting measures solely predicated on the explosion characteristics of pure fine dust. This, perhaps, constitutes a significant contributory factor to the heightened incidence of dust explosion incidents involving large particulate matter.

In conclusion, much of the research being done now focuses on flammable typical tiny dust particles according to established testing procedures, with little focus on the possible effects of large particles on the ignition and spread of fine particles. Based on this, this project plans to use a dust cloud combustion pipeline and a dust cloud MIT experimental platform to investigate the impact mechanism of coarse and fine dust mixing on the ignition and propagation of dust clouds in vertical pipelines. The primary focus of this essay is the mechanism by which debris affect dust ignition and spread. In process industries by providing strategies for their avoidance and management during real vertical mechanical transportation operations.

Experimental samples

Due to their prevalence in the grain manufacturing and processing process and the frequent occurrence of dust explosion mishaps during actual production, debris and rice husk are chosen as experimental materials (Amyotte, 2014; Amyotte et al., 2007; Zhang et al., 2024a). The sample of rice husk powder had a median diameter D_{50} of $17.56 \mu\text{m}$ (Fig. 1a). The debris sample is shown in Fig. 1b, following sieving, the particle size of each crushed brown rice is approximately uniform, resulting in

Impression of flame propagation of rice husk powder by debris

Rice husk powder was uniformly blended with various amounts of debris and ignited to examine the impact of big particles of debris on the flame propagation of rice husk dust. Fig. 4bcd shows the time series of flame propagation when the dust cloud concentration is 500 g/m^3 and the proportion of debris is 10 %, 30 %, and 70 %. Additionally, images of flame propagation in pure rice husk dust are provided for comparison (Fig. 4a). At $t = 10 \text{ ms}$, the pure rice husk dust cloud is ignited, and the

Discussion

Existing research indicates that a substantial presence of large particles exhibits an inerting effect akin to inert materials, mitigating the risks and consequences of dust explosions. On the other hand, little amounts of big particles added to mixed dust clouds improve their dispersion, resulting in a finer effective particle size, thus facilitating ignition and flame propagation. Fig. 19 illustrates the dispersion and ignition propagation mechanism of mixed dust clouds when debris are

Conclusion

The current work examines how different amounts of large particles additives affect rice husk dust cloud MIT and flame propagation behavior. It is observed that the alteration in dispersion characteristics induced by large particles similarly exacerbates the combustion of mixed dust clouds and escalates the FSV, a phenomenon evident across various concentration ranges.

Moreover, the presence of debris enhances the dispersibility of dust clouds, reducing the efficient particle size and

CRedit authorship contribution statement

Jinglin Zhang: Writing – original draft. **Gang Li:** Formal analysis. **Zhenguo Du:** Data curation. **Shikai Bao:** Funding acquisition. **Chang Li:** Conceptualization. **Zhiyang Zhang:** Methodology. **Chunmiao Yuan:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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